

An Anatomy of Precision: The Epsilon–Delta Rigor of Limits and Continuity

A Mathematical Essay on Quantifiers, Inequalities, and Edge Cases

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Abstract

The Cauchy–Weierstrass (ϵ, δ) -definitions of function limits and continuity are among the basic tools of analysis. While introductory calculus presents these definitions as standard technical formalisms, their small details matter: the quantifiers, inequalities, and exclusions are not interchangeable. This essay examines the definitions of function limits and continuity, with special attention to why $0 < |x - a|$ is required for limits but absent in continuity, why strict inequalities are chosen over non-strict inequalities in quantifiers, and what goes wrong if these conditions are modified or relaxed.

Contents

1	Introduction	2
2	A Prototype: Equality by Arbitrary Precision	2
3	The Formal Definitions	3
4	Excluding Zero: The Role of $0 < x - a$	3
4.1	Why Exclude $x = a$ in Limits?	3
4.2	Why Include $x = a$ in Continuity?	4
5	Strict vs. Non-Strict Inequalities in Quantifiers	4
5.1	Can we replace $ f(x) - L < \epsilon$ with $ f(x) - L \leq \epsilon$?	4
5.2	Can we replace $0 < x - a < \delta$ with $0 < x - a \leq \delta$?	5
5.3	What if we allow $\epsilon = 0$ ($\epsilon \geq 0$)?	5
5.4	What if we allow $\delta = 0$ ($\delta \geq 0$)?	5
6	Summary Matrix of Variant Definitions	5
7	Conclusion	6

1 Introduction

In early calculus, a limit $\lim_{x \rightarrow a} f(x) = L$ is often intuitively described as: “as x gets arbitrarily close to a , $f(x)$ gets arbitrarily close to L .” While visually appealing, this heuristic relies on kinematic concepts (“gets close to”, “approaches”) that invite logical paradoxes and ambiguity.

During the 19th century, Augustin-Louis Cauchy and Karl Weierstrass arithmeticized analysis by replacing dynamic motion with static logical quantifiers. The modern definitions convert topological proximity into precise real-number inequalities. However, it is easy to take the structural details of these definitions for granted. Why do we write $0 < |x - a| < \delta$ instead of $|x - a| < \delta$? Why do we require $\epsilon > 0$ strictly? What breaks if we replace $< \epsilon$ with $\leq \epsilon$?

This essay analyzes these subtleties and shows that some small-looking changes alter the meaning of the definitions.

2 A Prototype: Equality by Arbitrary Precision

Before looking at limits, it is useful to isolate a simpler logical pattern that already contains the heart of (ϵ, δ) reasoning. The pattern is this: if a nonnegative quantity can be made smaller than every positive tolerance, then that quantity must be zero.

Theorem 2.1 (Equality by Arbitrary Precision). *For any real numbers $a, b \in \mathbb{R}$,*

$$a = b \iff \forall \epsilon > 0, |a - b| < \epsilon.$$

Proof. We prove both directions.

Direction 1 ($a = b \implies \forall \epsilon > 0, |a - b| < \epsilon$): Assume $a = b$. Then $a - b = 0$, so $|a - b| = 0$. Let $\epsilon > 0$ be arbitrary. Since $0 < \epsilon$, we have $|a - b| = 0 < \epsilon$. Therefore

$$\forall \epsilon > 0, |a - b| < \epsilon.$$

Direction 2 ($\forall \epsilon > 0, |a - b| < \epsilon \implies a = b$): Assume

$$\forall \epsilon > 0, |a - b| < \epsilon.$$

We will show that $a = b$. Suppose, for contradiction, that $a \neq b$. Then $|a - b| > 0$. Choose

$$\epsilon_0 = \frac{|a - b|}{2}.$$

Since $|a - b| > 0$, we have $\epsilon_0 > 0$. By the assumed statement, applied to this particular positive tolerance ϵ_0 , we get

$$|a - b| < \epsilon_0 = \frac{|a - b|}{2}.$$

But this is impossible for a positive number $|a - b|$: it would imply

$$\frac{|a - b|}{2} < 0.$$

This contradiction shows that our supposition $a \neq b$ was false. Hence $a = b$. □

This theorem explains why the phrase “for every $\epsilon > 0$ ” is so powerful. It does not merely say that $|a - b|$ is small. It says that no positive tolerance, however tiny, can still detect any separation between a and b . The only nonnegative distance with that property is zero.

The (ϵ, δ) -definition of a limit adds one extra layer to this prototype. Instead of directly proving that a fixed distance $|a - b|$ is smaller than every ϵ , it says:

$$\forall \epsilon > 0, \exists \delta > 0$$

such that whenever x is within the permitted δ -neighborhood of a , the output distance $|f(x) - L|$ is smaller than that chosen ϵ . In other words, the δ clause is a mechanism for forcing the output error below an arbitrary tolerance.

Thus a limit proof usually has the same skeleton as Theorem 2.1, but with a dependency inserted:

$$\text{given an arbitrary } \epsilon > 0, \quad \text{choose a suitable } \delta > 0, \quad \text{then prove } |f(x) - L| < \epsilon$$

for every x satisfying the domain and closeness conditions. The theorem above is the degenerate case where there is no input variable and no δ to choose: the relevant distance is already fixed.

3 The Formal Definitions

Let $D \subseteq \mathbb{R}$ be the domain of a real-valued function $f : D \rightarrow \mathbb{R}$, and let $a \in \mathbb{R}$ be a limit point (or accumulation point) of D .

Definition 3.1 (Limit of a Function). We say that the **limit of $f(x)$ as x approaches a is L** , denoted $\lim_{x \rightarrow a} f(x) = L$, if:

$$\forall \epsilon > 0, \exists \delta > 0, \forall x \in D, (0 < |x - a| < \delta \implies |f(x) - L| < \epsilon).$$

Now consider the definition of continuity at a point. Let $a \in D$.

Definition 3.2 (Continuity at a Point). We say that f is **continuous at a** if:

$$\forall \epsilon > 0, \exists \delta > 0, \forall x \in D, (|x - a| < \delta \implies |f(x) - f(a)| < \epsilon).$$

Comparing Definition 3.1 and Definition 3.2, two structural differences immediately emerge:

1. **The explicit value of L vs. $f(a)$:** In the limit definition, L is a candidate value not necessarily tied to $f(a)$ (indeed, $f(a)$ need not even be defined). In continuity, $f(a)$ is strictly required to exist, and $L = f(a)$.
2. **The lower bound $0 < |x - a|$:** The limit definition explicitly excludes $x = a$, whereas the definition of continuity includes $x = a$.

4 Excluding Zero: The Role of $0 < |x - a|$

The condition $0 < |x - a|$ is equivalent to stating $x \neq a$. Its inclusion in the definition of a limit, and its omission in the definition of continuity, is not accidental.

4.1 Why Exclude $x = a$ in Limits?

The limit $\lim_{x \rightarrow a} f(x)$ seeks to describe the *local behavior of f near a* , completely independent of the *value of f at a* .

Suppose we removed the constraint $0 < |x - a|$ and defined a "modified limit" L' using:

$$|x - a| < \delta \implies |f(x) - L'| < \epsilon.$$

If $x = a$ were allowed, then $|a - a| = 0 < \delta$ holds for every $\delta > 0$. The implication would then require:

$$|f(a) - L'| < \epsilon \quad \forall \epsilon > 0.$$

Since the only non-negative real number less than every positive ϵ is 0, this forces $|f(a) - L'| = 0$, meaning L' must equal $f(a)$.

This hypothetical alteration would immediately break standard examples in calculus:

- **Functions with removable discontinuities would have no limits.** Consider $f(x) = \frac{x^2-1}{x-1}$ for $x \neq 1$, with $f(1) = 5$. Intuitively and standardly, $\lim_{x \rightarrow 1} f(x) = 2$. However, under the modified definition, no limit could exist because $f(1) = 5 \neq 2$.
- **Functions undefined at a could not have limits.** The standard derivative definition is:

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}.$$

The difference quotient $g(h) = \frac{f(a+h)-f(a)}{h}$ is explicitly undefined at $h = 0$. If $h = 0$ were required in the limit quantifier, no derivative could ever be evaluated!

Thus, excluding $x = a$ decouples local trend from point behavior.

4.2 Why Include $x = a$ in Continuity?

For continuity, the situation is reversed. Continuity asserts that the local trend *matches* the point value.

If we evaluate the continuity condition at $x = a$:

$$|a - a| < \delta \implies |f(a) - f(a)| < \epsilon \iff 0 < \delta \implies 0 < \epsilon.$$

For $x = a$, the statement $|f(a) - f(a)| < \epsilon$ reduces to $0 < \epsilon$, which is vacuously true for all $\epsilon > 0$. Thus, explicitly writing $0 < |x - a| < \delta$ or $|x - a| < \delta$ in the definition of continuity produces logically equivalent statements. Including $x = a$ simply simplifies the notation and emphasizes that $f(a)$ must be defined.

5 Strict vs. Non-Strict Inequalities in Quantifiers

A subtle point of confusion for students is why we use strict inequalities ($\epsilon > 0$, $\delta > 0$, $|x - a| < \delta$, $|f(x) - L| < \epsilon$) rather than weak inequalities ($\geq, \leq$). Does changing strict inequalities to weak ones change the mathematical meaning?

Let us test all permutations systematically.

5.1 Can we replace $|f(x) - L| < \epsilon$ with $|f(x) - L| \leq \epsilon$?

Theorem 5.1. *The definition of limit with $|f(x) - L| \leq \epsilon$ is equivalent to Definition 3.1.*

Proof. Let Definition A be the standard definition using $< \epsilon$, and Definition B be the modified definition using $\leq \epsilon$.

Direction 1 (A $\implies$ B): Assume $\lim_{x \rightarrow a} f(x) = L$ under Definition A. Given $\epsilon > 0$, by Definition A there exists $\delta > 0$ such that $0 < |x - a| < \delta \implies |f(x) - L| < \epsilon$. Since $|f(x) - L| < \epsilon$ implies $|f(x) - L| \leq \epsilon$, Definition B holds with the same δ .

Direction 2 (B $\implies$ A): Assume Definition B holds. Given $\epsilon > 0$, choose $\epsilon' = \frac{\epsilon}{2} > 0$. By Definition B, there exists $\delta > 0$ such that $0 < |x - a| < \delta \implies |f(x) - L| \leq \epsilon' = \frac{\epsilon}{2}$. Since $\frac{\epsilon}{2} < \epsilon$, we have $|f(x) - L| < \epsilon$. Thus Definition A holds. $\square$

Hence, using $\leq \epsilon$ instead of $< \epsilon$ changes nothing mathematically. The strict inequality $< \epsilon$ is preferred purely for aesthetic symmetry with $|x - a| < \delta$.

5.2 Can we replace $0 < |x - a| < \delta$ with $0 < |x - a| \leq \delta$?

Theorem 5.2. *The definition of limit with $0 < |x - a| \leq \delta$ is equivalent to Definition 3.1.*

Proof. If a $\delta_1 > 0$ satisfies the strict inequality condition, taking $\delta_2 = \frac{\delta_1}{2} > 0$ satisfies the weak inequality condition because $0 < |x - a| \leq \delta_2 \implies 0 < |x - a| < \delta_1$. Conversely, if δ_1 works for weak inequality, it automatically works for strict inequality since $0 < |x - a| < \delta_1 \implies 0 < |x - a| \leq \delta_1$. $\square$

5.3 What if we allow $\epsilon = 0$ ($\epsilon \geq 0$)?

What happens if we modify the quantifier from $\forall \epsilon > 0$ to $\forall \epsilon \geq 0$?

Suppose we require:

$$\forall \epsilon \geq 0, \exists \delta > 0, 0 < |x - a| < \delta \implies |f(x) - L| \leq \epsilon.$$

Setting $\epsilon = 0$, the condition demands that there exists $\delta > 0$ such that for all x in the punctured neighborhood $0 < |x - a| < \delta$:

$$|f(x) - L| \leq 0 \implies f(x) = L.$$

This forces $f(x)$ to be *strictly constant* in an open punctured neighborhood around a .

Thus, allowing $\epsilon = 0$ makes the definition far too strong. Non-constant functions like $f(x) = x$ would fail to have a limit anywhere! The strict positivity $\epsilon > 0$ is what allows $f(x)$ to approximate L without forcing immediate equality.

5.4 What if we allow $\delta = 0$ ($\delta \geq 0$)?

Suppose the definition allowed choosing $\delta = 0$. If $\delta = 0$, the condition $0 < |x - a| < 0$ is a contradiction—no real number x can satisfy it. The antecedent of the implication:

$$0 < |x - a| < 0 \implies |f(x) - L| < \epsilon$$

becomes *vacuously true* for *every* real number L and every function f .

If $\delta = 0$ were allowed, every function would have *every* real number as its limit at every point, so limits would no longer be unique. Hence, δ must be strictly positive ($\delta > 0$).

6 Summary Matrix of Variant Definitions

To synthesize these logical dependencies, the table below summarizes the effect of altering individual components of the (ϵ, δ) -definition for $\lim_{x \rightarrow a} f(x) = L$:

Modification	Mathematical Consequence	Status
Replace $< \epsilon$ with $\leq \epsilon$	Equivalent.	Valid alternative.
Replace $< \delta$ with $\leq \delta$	Equivalent.	Valid alternative.
Remove $0 < x - a $	Forces $L = f(a)$. Breaks derivatives.	Invalid.
Allow $\epsilon \geq 0$	Forces $f(x) = L$ locally (constant function).	Invalid.
Allow $\delta \geq 0$	Vacuously true for all L . Limits not unique.	Invalid.
Swap quantifiers: $\exists \delta \forall \epsilon$	Forces $f(x) = L$ identically near a .	Invalid.

Table 1: Impact of small formal changes on the limit definition.

7 Conclusion

The (ϵ, δ) -definition of a limit is carefully tuned. The inclusion of $0 < |x - a|$ separates the limit from the function's value at the point, which is what makes removable discontinuities and derivative quotients possible. The strict positivity of ϵ and δ leaves room for approximation without collapsing the definition into exact equality or vacuous truth. Choices that can look like minor notational preferences are often doing real logical work.